Codex Critique of Claude — 2026-11-08 cycle

Step 2: independent, stock-by-stock cross-critique. Prepared November 9, 2026, America/Chicago; completed after the November 9 regular-session close. All comparison prices are that close, in USD. The terminal date is December 31, 2030, 4.1424 years later. The filename denotes the analysis cycle, not the market-data cutoff.

Inputs: both November 8 deep dives, including Codex's subsequent framework audit, and the current Step 1 and Step 2 prompts. I did not read Claude's Step 2 critique. This is not a merged portfolio, a replacement deep dive, or Step 3 synthesis. The original reports and live deployment documents are unchanged. Forecasts below are my assumptions, not company guidance unless specifically identified; this is investment research, not a guarantee or trade instruction.

1. Verdict

• **Claude's best addition is Micron.** Its longer-duration customer commitments justify more durable earnings than my original memory normalization allowed. At the same November 9 price, my new MU estimate is 12.2% expected CAGR, versus Coherent's 8.8%; AI-2027 returns are 20.1% versus 14.6%. I replace COHR's entire 8% with MU. This does not require accepting Claude's $130 terminal EPS, embedded buybacks, or the idea that memory is no longer cyclical.

• **Claude was closer on ASML's operating opportunity.** My old earnings bridge understated the capacity expansion. ASML now beats my preferred excluded equipment challenger, Lam Research: 12.3% versus 8.4% expected CAGR. It nevertheless loses the last capital slot to MU under the required higher-AI-scenario-return tie-break.

• **Claude's weakest holding is Kratos.** The independently reconstructed common-earnings, dilution and investment bridge gives approximately −1.3% expected CAGR, versus Claude's repriced 9.9%. A good procurement story does not establish a good common-stock return. The proposed defense/materials/biotech sleeve also is not demonstrated catastrophe insurance.

• **Claude's caution on Broadcom and Coherent improves my underwriting.** I reduce terminal multiples and common-share economics and explicitly charge Broadcom's financing exposure. But contingent lease support is not the same as an immediately payable face-value debt obligation. The financing charge alone does not justify a full AVGO exit; the combined return comparison does.

• **My strongest retained disagreements are CLS, NVT and NRG.** Claude's zero weights leave substantial estimated return on the table after current guidance, organic growth, sale-related earnings loss, capital spending and debt are accounted for. I retain 12% each, not because these are mandatory bottleneck sleeves.

• **Verisk, which I underwrote but excluded and Claude never considered, now displaces Broadcom.** Its unchanged operating model, repriced to November 9, offers 15.0% expected CAGR versus AVGO's re-underwritten 12.1%. Its AI-2027 return is also slightly higher. The funding is AVGO 15% → VRSK 15%; this is not a concession to Claude or a token hedge.

• **Revised own allocation:** NVDA 15%, VRT 15%, VRSK 15%, CLS 12%, NVT 12%, NRG 12%, VST 11%, MU 8%. Checksum **100.0%**. Using identical new underwriting, expected CAGR improves from 17.3% to 17.9%, AI-2027 CAGR from 22.3% to 22.7%, and the simultaneous-bust terminal loss improves from 65.0% to 57.9%. The 100GW CAGR falls from 44.9% to 42.5%: I accept that cost because expected return is primary.

• **This is a conditional, fragile improvement, not a proven optimum.** A 20% reduction in Verisk's terminal P/Es can erase the expected-return advantage. Eight holdings still concentrate on common AI spending, power and financial-market risks. Step 3 must compare these revised estimates with the remaining previously underwritten universe at the same prices and assumptions; this bounded cross-critique does not certify that every non-contested challenger loses.

2. Head-to-Head: Every Stock Where the Portfolios Differ

Comparison rules and provenance

The union of the two proposed portfolios contains 19 stocks. Every one qualifies: it is held by only one model or differs by at least one percentage point. H01–H19 cover all 19; H20 adds the previously excluded candidate that wins capital in this critique. Zero means a proposed allocation of zero, not that a company was never researched.

The first two expected-CAGR columns preserve each original model's terminal wealth and probabilities, changing only price and the time remaining. The third uses my new underwriting. Claude's published rounded terminal values are retained as published; I do not manufacture extra precision or an unavailable point-in-time consensus series. NR means the original did not disclose a full numerical valuation, not zero expected return.

Neither original deep dive supplied the newly required named-scenario returns. Columns marked * are **new reconstructions made here**, not historical statements by either author. I apply the shared named paths described below to the original base economics. For earnings models, the stock-specific change in revenue × common margin / diluted shares, and the change in exit multiple, transport each original base to the named path. Original dividends are retained; explicit excess cash is scaled separately. For my power models, EBITDA, multiple, claims and shares are transported individually. Claude's VST uses its original FCF/share valuation, with the corresponding change in FCF/share and multiple. This holds the scenario loading constant while exposing original differences in the shareholder-value bridge.

† identifies weaker identification: where an original has terminal values but no operating bridge, I scale its base terminal wealth by the new named/base wealth ratio. Where the entire original valuation is absent, I show the new model as an explicitly labeled proxy. These are computable sensitivities, not recovered author forecasts. In particular, Claude's NVT/NRG and the four Codex-only candidates cannot be given genuine original-model returns from information that was never supplied. The adopted named returns appear separately in the outcome and operating-bridge tables; those direct models, not the proxies, determine allocation.

Row / ticker	Mine %	Claude %	Price + date	Expected CAGR: mine / theirs / new	AI-2027 CAGR: mine* / theirs*	100GW CAGR: mine* / theirs*	Input gap	Better call	Why
H01 NVDA	15	13	$178.67; 2026-11-09	25.9% / 17.9% / 21.9%	27.9% / 21.7%	56.1% / 48.8%	Base EPS 16.64 vs 19; P/E 25 vs 18; bull P/E 30 vs 22.	Codex weight; reduce my valuation	22x base/26x bull retains platform value without my old premium. Still strongest selected return. H01.
H02 AVGO	15	6	$319.48; 2026-11-09	19.0% / 12.2% / 12.1%	24.7% / 17.6%	48.1% / 39.7%	Base P/E 24 vs 20; margin/share count; financing reserve absent in mine.	Claude on valuation; neither weight	47% margin, 5.0bn shares, 20x and explicit credit reserve. VRSK wins new capital; S06.
H03 VRT	15	9	$200.47; 2026-11-09	21.0% / 15.0% / 16.0%	25.9% / 21.7%	55.5% / 50.3%	Similar base EPS; 26x vs 23x, much larger Codex bull.	Claude valuation; Codex weight	24x base, 19% common margin, lower bull. Still beats lower-return equipment alternatives.
H04 CLS	12	0	$222.92; 2026-11-09	21.3% / 11.9% / 19.2%	28.6% / 19.4%†	52.8% / 42.0%†	My base $513 vs their $378; their EPS/multiple bridge absent.	Codex	2027 issuer outlook supports growth beyond their terminal hurdle; dilution and cash needs trim my target. F05.
H05 NVT	12	0	$114.44; 2026-11-09	21.5% / NR / 19.4%	26.3% / 26.3%†	50.4% / 50.5%†	No Claude terminal underwriting; rejected mainly on smaller cooling share.	Codex	Electrical distribution economics and valuation beat ETN; cooling market share is not the entire business. S01.
H06 NRG	12	0	$97.40; 2026-11-09	21.6% / NR / 20.1%	25.3% / 24.1%†	36.1% / 35.0%†	No Claude terminal underwriting; retail/gas discount dismissed.	Codex, with a stricter debt bridge	Asset-sale earnings removed; proceeds not spent twice. Still beats CEG. S02.
H07 VST	11	9	$133.92; 2026-11-09	18.3% / 13.2% / 16.9%	21.3% / 18.0%	36.7% / 29.5%	My EV bridge vs their FCF/share; acquisition scope misread.	Codex inclusion; lower return than mine	No extra Cogentrix uplift beyond consolidated guide; retain growth-capex and debt charges. F03.
H08 COHR	8	5	$225.19; 2026-11-09	14.7% / 9.8% / 8.8%	24.2% / 17.1%	49.1% / 40.6%	24x vs 18x base P/E; cash conversion and dilution.	Claude caution; both overweight	New 18% margin, 230m shares, 20x. MU wins E and AI returns; exit, S05.
H09 TSM	0	14	$388.62; 2026-11-09	12.6% / 14.1% / 12.1%	16.4% / 16.6%	36.0% / 36.3%	Base EPS 26.83 vs 33; probabilities 25/50/25 vs 20/50/30.	Codex zero; improve my earnings case	Higher revenue, lower exit premium; all-in build cost is not foundry ASP. Taiwan remains a common risk.
H10 MU	0	9	$733.15; 2026-11-09	11.4% / 14.0% / 12.2%	16.7% / 18.3%	29.2% / 31.4%	My EPS 80 plus $150 cash vs their $130 with embedded buybacks.	Claude inclusion	Use $110 EPS, explicit cash, fixed shares, 10x. Beats COHR; wins near-ties on AI return.

Row / ticker	Mine %	Claude %	Price + date	Expected CAGR: mine / theirs / new	AI-2027 CAGR: mine* / theirs*	100GW CAGR: mine* / theirs*	Input gap	Better call	Why
H11 ASML	0	8	$1,513.25; 2026-11-09	7.0% / 13.0% / 12.3%	12.2% / 19.3%	28.9% / 37.1%	My base EPS 63.47 vs 88; capacity/margin/FX bridge.	Claude underwriting direction	New $85.65 base EPS; ASML beats my preferred WFE challenger. MU wins the last allocation slot.
H12 SNPS	0	6	$466.97; 2026-11-09	11.9% / 11.8% / 11.9%	15.1% / 16.5%	28.1% / 29.6%	Base EPS 24 vs 27; 30x vs 28x; similar expected values.	Codex zero	No decisive return edge. VRSK has stronger expected return; not a substitute for EDA technology.
H13 CEG	0	6	$276.09; 2026-11-09	10.1% / 8.8% / 10.4%	12.9% / 13.5%	29.1% / 29.9%	2030 earnings and Nov 9 update; nuclear premium.	Codex zero	Recognize higher guidance; price also rose. NRG/VST remain cheaper power returns.
H14 ETN	0	5	$359.30; 2026-11-09	13.5% / 10.6% / 12.6%	17.9% / 16.8%	36.1% / 35.0%	Similar base EPS; probability tails and quality multiple.	Codex NVT	ETN is credible quality; NVT offers about 6.8pp higher expected CAGR. S01.
H15 CRDO	0	3	$132.89; 2026-11-09	8.2% / 15.3% / 12.2%	17.6% / 25.2%	41.4% / 50.6%	My base EPS 6.36 vs 10.5; compensation and share dilution.	Claude improves my growth view; no add	New base EPS 8.35, after economic compensation; MU wins close expected-return tie on AI CAGR.
H16 PWR	0	2	$591.48; 2026-11-09	8.6% / 6.0% / 8.2%	14.1% / 13.2%	30.1% / 29.1%	Similar base EPS 32 vs 31; price and execution costs.	Codex zero	Strong infrastructure business, only 8.2% expected at current price; equipment wins.
H17 KTOS	0	2	$37.69; 2026-11-09	-2.0% / 9.9% / -1.3%	0.6% / 15.3%	12.4% / 28.8%	Base EPS 1.16 vs 1.9; owner margin, share count, bull probability.	Codex	Production investment and dilution prevent adjusted EPS from being owner cash. Weakest Claude holding.
H18 MP	0	2	$37.75; 2026-11-09	6.9% / 5.6% / 7.2%	12.6% / 10.6%	31.4% / 29.1%	Bear floor and state probabilities; subsidy/contract vs common return.	Codex zero	Qualification is progress; expected return still loses and no demonstrated crisis hedge.
H19 BNTX	0	1	$87.51; 2026-11-09	NR / 9.2% / 4.3%	7.9%† / 11.5%†	12.0%† / 15.7%†	No original Codex model; Claude $65 bear and $120 base total wealth.	Codex exclusion; new explicit model	Cash burns before 2030; lower pipeline SOTP. Not reliable pandemic insurance.
H20 VRSK	0	0	$200.60; 2026-11-09	15.0% / NR / 15.0%	17.4% / 17.4%†	19.9% / 19.9%†	Previously excluded by me; absent from Claude universe.	New Codex allocation	15.0% E vs AVGO 12.1%; AI 17.4% vs 17.0%. Own existing model now clears the opportunity cost.

Re-underwriting the material gaps

All original expected-CAGR differences above 3 percentage points are re-underwritten below, including LRCX, which enters the equipment comparison in §3. The selected inputs are not averages of the two estimates. A source can support revenue visibility without proving a particular 2030 multiple; the latter remains an explicit valuation judgment.

Stock	Repriced original gap, mine minus Claude	Input-level resolution	Why this is not an average
NVDA	+7.9pp	Base revenue $850bn, common margin 46%, diluted shares 23.5bn, EPS $16.64, 22×; bull 26× rather than my 30×.	Retain the original operating bridge; reduce the duration premium. Claude's lower multiple identifies a real risk, but its higher base EPS is not needed to justify my 15% weight.
AVGO	+6.8pp	Base $250bn revenue, 47% common margin, 5.0bn shares, $23.50 EPS, 20×; $13 dividends and −$2 credit reserve per share.	Choose a lower normalized common margin, explicit dilution and liability reserve; no averaging of target prices. Financing risk and a lower terminal premium act separately. F07.
VRT	+6.0pp	Base $30bn revenue, 19% common margin, 380m shares, $15 EPS, 24×. Bull revenue $45bn, margin 22%, 27×.	Similar base earnings did not explain my much larger tail. Reduce both the bull economics and exit premium; retain the weight only because the resulting return still wins capital.
CLS	+9.4pp	Base $65bn revenue, 6.5% common margin, 145m shares, $29.14 EPS, 16×.	Issuer 2027 outlook is the operating anchor; add dilution and cash-conversion discipline. Claude's $378 base wealth would imply $23.63 EPS at a matched 16×, but that is my diagnostic, not Claude's disclosed assumption. F05.
VST	+5.2pp	Base EBITDA $11bn, 9× EV/EBITDA, $20bn claims, 300m shares and $5 dividends; FCF before discretionary growth $5.5bn.	Value the consolidated company, charge debt, and do not add an unsupported acquisition increment to the post-close guide. F03.
COHR	+5.0pp	Base $20bn revenue, 18% common margin, 230m shares, $15.65 EPS, 20×.	Growth alone does not finance new facilities. Reduce common margin, increase dilution and lower the multiple together; current cash consumption is the evidence for greater discipline, not a claim that one quarter lasts forever. F17.
ASML	−6.0pp	Base €80bn revenue at an assumed $1.10/€, 36.5% common margin, 375m shares, $85.65 EPS, 28×.	Rebuild capacity and service revenue from units, assumed ASPs and installed-base activity; do not average $63.47 and $88 EPS. F09 and the bridge below.
CRDO	−7.1pp	Base $6bn revenue, 32% common margin after economic compensation, 230m shares, $8.35 EPS, 22×.	Strong sales growth raises my revenue case; actual compensation and dilution prevent importing adjusted EPS wholesale. The assumed margin improves from the current GAAP level but remains far below the current adjusted margin. F06.
KTOS	−11.9pp	Base $4.1bn revenue, 8% common margin, 240m shares, $1.37 EPS, 28×; 30/50/20 probabilities.	Require manufacturing investment and share dilution before assigning value to funded procurement. Revenue growth is not equivalent to distributable common earnings. F15.
LRCX	+9.2pp	Base $45bn revenue, 33% common margin, 1.20bn shares, $12.38 EPS, 22×.	Reduce my old revenue/margin/multiple combination for cyclicality and geographic exposure; do not assume an unverified future export restriction has already become law. F20.

NVT, NRG and BNTX require new numerical underwriting despite the absence of two comparable original estimates. MU's original gap is under 3pp, but it must also be rebuilt because it changes an allocation. TSM's probability and unit interpretation, and CEG's newly released guidance, likewise affect the head-to-head verdict even without a 3pp trigger.

Complete current-price outcome ledger

Terminal wealth includes terminal common-equity value plus cumulative nominal dividends; no dividend reinvestment, taxes or trading costs are assumed. Expected wealth is calculated first, then annualized. Portfolio returns use weighted wealth ratios, not an average of stock CAGRs. Values are rounded for display; the companion JSON retains calculation precision.

E[W] = Σ(probability × terminal wealth); Expected CAGR = (E[W] / current price)^(1 / 4.1424) − 1.

The default company financial-state probabilities are 25/50/25. KTOS, MP, BNTX and SNDK use 30/50/20 because dilution, project/clinical outcomes or commodity normalization create a less favorable distribution. These are subjective company-financial judgments, not claimed frequencies or the probabilities of particular AGI dates. They are not obtained by counting how many historical AI predictions came true.

Ticker	Nov 9 price	Bear / base / bull wealth $	Probabilities %	Expected wealth $	Expected CAGR	AI-2027 CAGR	100GW CAGR
NVDA	$178.67	78.667 / 372.043 / 799.304	25 / 50 / 25	$405.51	21.9%	24.1%	51.5%
AVGO	$319.48	104.941 / 481 / 985.697	25 / 50 / 25	$513.16	12.1%	17.0%	39.0%
VRT	$200.47	50.231 / 362 / 706.421	25 / 50 / 25	$370.16	16.0%	22.0%	50.7%
CLS	$222.92	60.345 / 466.207 / 855	25 / 50 / 25	$461.94	19.2%	25.6%	49.3%
NVT	$114.44	56.988 / 237.75 / 420.625	25 / 50 / 25	$238.28	19.4%	26.3%	50.5%
NRG	$97.40	41.333 / 209 / 373.889	25 / 50 / 25	$208.31	20.1%	24.1%	35.0%
VST	$133.92	51.059 / 268.333 / 433.586	25 / 50 / 25	$255.33	16.9%	21.3%	36.8%
COHR	$225.19	34.087 / 313.043 / 616	25 / 50 / 25	$319.04	8.8%	14.6%	37.7%
TSM	$388.62	141.792 / 651.615 / 1,051.699	25 / 50 / 25	$624.18	12.1%	17.0%	36.8%
MU	$733.15	264 / 1,258 / 1,949.5	25 / 50 / 25	$1,182.38	12.2%	20.1%	33.1%
ASML	$1,513.25	528.701 / 2,438.293 / 4,383.333	25 / 50 / 25	$2,447.16	12.3%	18.5%	36.3%
SNPS	$466.97	201.951 / 720 / 1,335.122	25 / 50 / 25	$744.27	11.9%	15.1%	28.1%
CEG	$276.09	117.889 / 405.714 / 735	25 / 50 / 25	$416.08	10.4%	14.1%	30.5%
ETN	$359.30	178.615 / 567.684 / 1,031.789	25 / 50 / 25	$586.44	12.6%	16.7%	34.9%
CRDO	$132.89	31.418 / 183.652 / 458.449	25 / 50 / 25	$214.29	12.2%	18.4%	42.5%
PWR	$591.48	194 / 803 / 1,473.333	25 / 50 / 25	$818.33	8.2%	14.1%	30.1%
KTOS	$37.69	4.109 / 38.267 / 76.8	30 / 50 / 20	$35.73	-1.3%	4.7%	17.0%
MP	$37.75	0.96 / 50.286 / 124.727	30 / 50 / 20	$50.38	7.2%	12.6%	31.4%
BNTX	$87.51	45 / 105 / 190	30 / 50 / 20	$104.00	4.3%	7.9%	12.0%
SNDK	$886.51	390 / 1,372.857 / 2,446.667	30 / 50 / 20	$1,292.76	9.5%	18.5%	31.8%
RELX	$40.02	28 / 65.617 / 109.26	25 / 50 / 25	$67.12	13.3%	15.1%	18.9%

Ticker	Nov 9 price	Bear / base / bull wealth $	Probabilities %	Expected wealth $	Expected CAGR	AI-2027 CAGR	100GW CAGR
VRSK	$200.60	138.769 / 355.5 / 581.932	25 / 50 / 25	$357.93	15.0%	17.4%	19.9%
MSI	$527.97	252.471 / 703.25 / 1,199.875	25 / 50 / 25	$714.71	7.6%	10.8%	15.1%
GEN	$29.80	13.953 / 35.549 / 62.5	25 / 50 / 25	$36.89	5.3%	6.2%	7.1%
LRCX	$221.31	81.8 / 279.25 / 595.102	25 / 50 / 25	$308.85	8.4%	14.1%	31.7%

Auditable operating inputs

All triples are bear / base / bull. Revenue and diluted shares are billions; margins are normalized after-tax earnings attributable to common shareholders **after economic stock compensation and interest**. They are not headline adjusted operating margins. An earnings multiple is applied after that charge. Explicit cash is incremental net distributable cash not already paid out or used for the modeled buybacks; a negative number is a credit-loss reserve. Debt is not subtracted a second time from an already levered earnings valuation. Conversely, excess-cash additions must exclude earnings from that cash to avoid double counting.

Ticker	Revenue $bn bear / base / bull	Common margin %	Diluted shares bn	Exit P/E	Dividends $/share	Excess cash / credit reserve $/share
NVDA	350 / 850 / 1,400	32 / 46 / 50	24 / 23.5 / 23	16 / 22 / 26	4 / 6 / 8	0 / 0 / 0
AVGO	100 / 250 / 400	35 / 47 / 50	5.1 / 5 / 4.95	15 / 20 / 24	10 / 13 / 16	-8 / -2 / 0
VRT	12 / 30 / 45	10 / 19 / 22	0.39 / 0.38 / 0.38	16 / 24 / 27	1 / 2 / 3	0 / 0 / 0
CLS	25 / 65 / 95	3.5 / 6.5 / 7.5	0.145 / 0.145 / 0.15	10 / 16 / 18	0 / 0 / 0	0 / 0 / 0
NVT	5.5 / 10 / 14	10 / 17 / 19	0.163 / 0.16 / 0.16	16 / 22 / 25	3 / 4 / 5	0 / 0 / 0
COHR	7 / 20 / 28	8 / 18 / 22	0.23 / 0.23 / 0.24	14 / 20 / 24	0 / 0 / 0	0 / 0 / 0
TSM	140 / 340 / 450	32 / 46 / 47	5.3 / 5.2 / 5.15	15 / 21 / 25	15 / 20 / 25	0 / 0 / 0
MU	120 / 300 / 450	20 / 44 / 50	1.2 / 1.2 / 1.2	11 / 10 / 9	4 / 8 / 12	40 / 150 / 250
ASML	40 / 88 / 130	24 / 36.5 / 40	0.385 / 0.375 / 0.36	20 / 28 / 30	30 / 40 / 50	0 / 0 / 0
SNPS	10 / 16 / 23	18 / 30 / 34	0.205 / 0.2 / 0.205	23 / 30 / 35	0 / 0 / 0	0 / 0 / 0
CEG	35 / 53 / 68	8 / 13 / 15	0.36 / 0.35 / 0.34	14 / 20 / 24	9 / 12 / 15	0 / 0 / 0
ETN	29 / 48 / 65	12 / 18 / 21	0.39 / 0.38 / 0.38	18 / 24 / 28	18 / 22 / 26	0 / 0 / 0
CRDO	2.4 / 6 / 12	18 / 32 / 36	0.22 / 0.23 / 0.245	16 / 22 / 26	0 / 0 / 0	0 / 0 / 0
PWR	32 / 60 / 80	5 / 8 / 9.5	0.15 / 0.15 / 0.15	18 / 25 / 29	2 / 3 / 4	0 / 0 / 0
KTOS	2.1 / 4.1 / 6	2.5 / 8 / 10	0.23 / 0.24 / 0.25	18 / 28 / 32	0 / 0 / 0	0 / 0 / 0
MP	0.6 / 2 / 3.5	2 / 22 / 28	0.2 / 0.21 / 0.22	16 / 24 / 28	0 / 0 / 0	0 / 0 / 0
SNDK	30 / 55 / 85	20 / 36 / 45	0.15 / 0.14 / 0.135	9 / 9 / 8	0 / 0 / 0	30 / 100 / 180
RELX	12 / 17 / 22	20 / 26 / 29	1.8 / 1.75 / 1.73	18 / 24 / 28	4 / 5 / 6	0 / 0 / 0
VRSK	3.4 / 4.5 / 5.7	25 / 33 / 37	0.13 / 0.12 / 0.118	20 / 28 / 32	8 / 9 / 10	0 / 0 / 0
MSI	13 / 19 / 25	16 / 22 / 25	0.17 / 0.16 / 0.16	19 / 26 / 30	20 / 24 / 28	0 / 0 / 0
GEN	4.5 / 6 / 7.5	17 / 24 / 28	0.64 / 0.61 / 0.6	10 / 14 / 17	2 / 2.5 / 3	0 / 0 / 0

Ticker	Revenue $bn bear / base / bull	Common margin %	Diluted shares bn	Exit P/E	Dividends $/share	Excess cash / credit reserve $/share
LRCX	25 / 45 / 70	24 / 33 / 38	1.25 / 1.2 / 1.18	16 / 22 / 26	5 / 7 / 9	0 / 0 / 0

NRG and VST use enterprise value because acquisitions, financing claims and capital intensity make an EPS-only shortcut less reliable. Claims include modeled net debt and other financing obligations. Common wealth = max(0, EBITDA × EV multiple − claims) / diluted shares + dividends. FCF is an underwriting cross-check, **not another asset added to EV**.

Ticker	EBITDA $bn	EV/EBITDA	Net financial claims $bn	Diluted shares bn	Dividends $/share	Owner cash before discretionary growth $bn
NRG	4 / 6.5 / 8.5	6.5 / 8 / 9	19 / 14 / 11	0.21 / 0.19 / 0.18	8 / 9 / 10	1.8 / 3.5 / 4.7
VST	6 / 11 / 14	7 / 9 / 10	26 / 20 / 16	0.34 / 0.3 / 0.29	4 / 5 / 6	2.2 / 5.5 / 7.5

BNTX uses a net-cash-plus-pipeline SOTP: bear $6bn + $5.25bn; base $10bn + $16.25bn; bull $12bn + $35.5bn, each divided by 250m shares. These are cash remaining **after modeled burn**, plus risk-adjusted pipeline values; an upfront payment already in cash is not added again. The result is $45 / $105 / $190. Clinical probabilities are embedded in the pipeline estimates; the external 30/50/20 states describe different aggregate outcomes rather than reapplying the same trial probability mechanically.

Funding and valuation checks that matter to the decisions

Decision	Cash, capacity or valuation bridge	Failure condition
Retain NRG 12%	Starting net financial claims are approximately $22.8bn, not the $14bn terminal assumption. My illustrative 2027–30 post-sale FCF-before-growth path is $2.7/$3.0/$3.2/$3.5bn, totaling $12.4bn, plus $4bn sale proceeds. Roughly $8.8bn goes to net debt reduction, $1.7bn to dividends, $3bn to future buybacks and $2.9bn to discretionary growth. Twenty million shares repurchased at an assumed average $150 support a move from about 210m to 190m. These are my funding assumptions, not an issuer allocation promise.	The sale must close; the sold assets' earnings must disappear. I reduce the old $7bn base EBITDA to $6.5bn, assuming a $0.5bn contribution sold. The actual asset EBITDA was not disclosed in the cited release: a $0.3–0.8bn sensitivity is needed. More self-funded new-build spending, higher average repurchase prices, weaker retail cash or failed debt reduction lowers common value. Sale proceeds are spent once.
Add MU 8%	Base EPS is $300bn × 44% / 1.2bn = $110; 10× plus $150 net cash/share and $8 dividends gives $1,258. The cash assumption is $180bn aggregate, compared with approximately $55.6bn current cash/investments after debt and customer-deposit claims. It therefore requires roughly $124bn further net retention, after investment, distributions and deposit repayment. Shares remain fixed: I do not also spend the same cash on buybacks.	Floor/ceiling contracts do not lock in today's margin. Base earnings are deliberately below the next-quarter annualized adjusted EPS run-rate. Reject the model if normal-capex cash retention cannot fund the explicit terminal balance or if qualification advantage is commoditized.
Raise ASML's estimate, retain zero weight	My €80bn base revenue consists of 120 low-NA tools × €250m = €30bn; 20 high-NA × €450m = €9bn; other systems/DUV €23bn; services €18bn. At $1.10/€, that is $88bn. These unit counts and ASPs are my 2030 assumptions, not announced orders or company guidance. The 140 combined EUV units require continued supply-chain expansion.	Production capacity is not accepted/recognized sales. A lower service mix, weaker unit realization, delayed high-NA adoption, FX below the assumption or lower net margins defeats the $85.65 EPS bridge. A 28× exit multiple still assumes a valuable durable franchise.
Add VRSK 15%	The original model is unchanged: base revenue $4.5bn, 33% common margin, 120m shares, $12.375 EPS, 28×, plus $9 dividends. From the current revenue-guide midpoint, that is about 8.8% annual sales growth through 2030. The share reduction from roughly 128m is modest relative to recent buybacks, but the recent buyback pace is not extrapolated. A 90% owner-cash conversion assumption yields about $1.34bn at the terminal base.	Four-year owner cash of roughly $4–5bn must cover about $2.1bn repurchases, $1.1bn dividends and $0.7bn debt reduction; there is limited room for expensive acquisitions. The net-margin improvement from the current GAAP level is material. Data substitution, slower licensed growth or a lower exit multiple can invalidate the position.
Reduce AVGO to zero	Base operating equity value is $470/share, plus $13 distributions, less a $2/share contingent-credit reserve. At 5bn shares, that reserve is $10bn; the bear reserve is about $41bn. Neither is a forecast that the contractual maximum will be paid. The 20× multiple and 47% margin do more valuation work than the reserve alone.	If high growth and cash conversion support the old premium without additional financing risk, AVGO can beat VRSK. Conversely, credit losses can be correlated with a chip-demand bust; a reserve is not a complete legal or counterparty model.
Retain VST 11%, reduce COHR to zero	Power FCF before growth must fund growth, debt service/reduction, repurchases and distributions. Coherent's planned capacity expansion likewise cannot be treated as free merely because adjusted EPS rises. Both models use more conservative common-share economics than the original EPS headlines.	Underfunded expansion raises claims or shares. My power terminal claims and Coherent's 230m base shares are assumptions to test, not guaranteed management outcomes.

These are explicit consistency checks, not full annual cash-flow statements. Terminal earnings models remain less reliable where growth investment, customer deposits or stock compensation dominate cash generation. That limitation lowers confidence in MU/CRDO/COHR and in the far-right 100GW cases; it is not permission to count the same financing twice.

Named scenarios: physical paths before stock returns

AI-2027-like commercial path: 50% planning probability; closest to base. This is the most likely broad capability/demand family, not a 50% assertion that superintelligence arrives in calendar 2027. I use the authors' updated timing, not a claim that their original dramatic sequence has already been validated. The other broad families are slower/financing-constrained expansion, 25%, and faster/heavier expansion, 25%. The exact operating outcome at any company can still be bear, base or bull within a macro family. The company distributions are judgmental marginals, not a fitted joint simulation; the named point paths are sensitivities and are **not added again to expected wealth**.

S denotes a published scenario input, not an observed fact. A denotes my reconciliation or extension. The April 2025 published compute forecast, the older-title spreadsheet and the later AI-2040 model are different vintages; combining their headline outputs without a stock/flow reconciliation would be false precision. Source interpretation is audited in F10–F14 below and in the earlier local forecast audit.

AI-2027 planning assumption	2026	2027	2028	2029	2030
Annual worldwide AI capex, $tn	0.6 S	1.0 S	2.4 A, informed by later compute model	3.2 A	4.2 A
End-year global H100-equivalent compute stock, millions	40 S	100 S	245 A	470 A	825 A
Gross new H100-equivalent shipments, millions	26 A	65 A	155 A	245 A	395 A
Retirements, H100-equivalent millions	5 A	5 A	10 A	20 A	40 A
Physical accelerator packages, millions	12 A	21 A	35 A	50 A	70 A
HBM per package, GB	256 A	384 A	512 A	768 A	1,024 A
Implied shipped HBM, billion GB	3.07 A	8.06 A	17.92 A	38.40 A	71.68 A
Assumed realized HBM $/GB	20 A	15 A	12 A	9 A	7 A
Implied HBM revenue, $bn	61 A	121 A	215 A	346 A	502 A
Leading lab annual recognized revenue, $bn	50 A	140 S	350 A	600 A	1,000 A
Global peak AI electrical demand, GW	29 S	62 S	100 A	150 A	230 A
Capability assumption, A except updated source timing	Longer coding/research assistance	Meaningful early-arrival tail for automated coding	Material research automation	Broader expert automation; possible much faster takeoff	Widespread automation; capture and ownership still uncertain

The stock roll-forward uses a 19m opening compute stock in 2026: 19 + 26 − 5 = 40; then 40 + 65 − 5 = 100, and so on. This is a transparent constructed series, not a claim that all forecasts used those same retirements. Physical packages are not H100 equivalents. The early package counts are rounded anchors from the workbook; later package performance, memory density and pricing are my assumptions. The HBM dollar row is therefore **not** the workbook's published TAM or a verified current industry estimate. It is an internally calculable memory-demand sensitivity. The later capex path deliberately permits commercially funded expansion rather than imposing AI-2040's assumed 2029 pause.

For MU's AI case, a roughly 20% share of that modeled 2030 HBM revenue would supply about $100bn of its $360bn revenue assumption; the remaining approximately $260bn requires conventional DRAM/NAND and other memory demand. That is a demanding market-share and industry-revenue bridge, not an assertion that Micron owns 20% today. It helps explain why I retain a cyclical multiple instead of valuing memory like a software monopoly.

100GW single-cluster path: 10% planning probability; closest to bull. It is a subset of the faster/heavier family, not an additional 10 percentage points of expected-value probability. The defining constraint is one largest training cluster approaching 100GW in 2030. Twenty to forty separate campuses totaling 100GW do not satisfy it. The older reports predate the prompt's formal named-return requirements; this is a corrected new sensitivity, not a criticism that either author omitted a requirement that did not yet exist.

100GW path assumption	2026	2027	2028	2029	2030
Largest training cluster, GW	1 S	3.2 A	10 S	32 A	100 S
Largest cluster, million legacy H100 equivalents	1 S	3.2 A	10 S	32 A	100 S
Largest cluster construction scale	Tens of $bn S	High tens to $100bn A	Hundreds of $bn S	Several hundred $bn A	$1tn+ S
Worldwide annual AI investment, $tn	0.5 S	1.0 S, described as plausible	2 S	4 A	8 S
Annual accelerator shipments, million H100 equivalents	30 A within source's tens	55 A	100 S order of magnitude	200 A	400 A within source's hundreds
Worldwide AI electricity relative to then-current US electricity	5% S	Intermediate A	20% S	Intermediate A	About 100% S
AI wafer demand relative to then-current leading-edge TSMC capacity	25% S	Intermediate A	100% S	Intermediate A	400% S

These are the source's back-of-the-envelope equivalences, not modern benchmark-normalized hardware specifications. I anchor this path primarily to physical power and investment, rather than pretending its legacy H100 measure is identical to every later compute model. A 100GW continuous load alone is 876TWh/year before any additional overhead if 100GW denotes IT power; at 85% utilization it is about 745TWh. This is an order-of-magnitude infrastructure requirement, not a minor extrapolation from a national capacity announcement.

Distance from the path at the cutoff: the reviewed supplier results support very rapid deployment and spending, but I did not obtain a verified inventory of energized, coherently networked training clusters proving that the largest-cluster milestones have been achieved. Announced GW, equipment shipments, reserved interconnection and operating power are different quantities. The earlier forecast audit contains completed-period shipment evidence; it does not establish November operating utilization. The 2030 requirement is still approximately 100 times the source's 2026 single-cluster power milestone, while annual worldwide investment rises sixteenfold from its 2026 marker. Treating the destination as already financed or permitted would be unjustified.

Binding order and value capture: near-term interconnection, site permission, transformers/switchgear and firm generation bind delivery; qualified memory, packaging and lithography constrain equipment expansion; financing can stop deployment before either physical limit is reached. NVT/VRT and existing generators can capture some near-term urgency. MU and ASML gain if memory and tools remain scarce. But commissions, governments and large customers can redirect scarcity rents, while new supply and substitution reduce them. A global shortage is not evidence that any particular supplier receives all the surplus.

The named stock bridges below translate these paths into revenue or EBITDA, normalized margin, share count, claims and exit valuation. They are not blanket percentage uplifts. The 100GW case explicitly adds dilution/claims to capital-intensive names and often compresses margins despite higher sales. My ASML 100GW sales require capacity expansion well beyond the nearer-term tool plan; my MU 100GW revenue requires an enormous wider memory market, not HBM alone. Those are conditions of a low-probability scenario, not current orders. Upstream and downstream supplier revenues must not be summed as though they were independent additions to global capex.

R = revenue $bn; m = common net margin; S = diluted shares bn; D = cumulative dividends/share; C = excess net cash or credit reserve/share. For power, EBITDA/claims/FCF are $bn. FCF remains a check, not an added asset.

Ticker	AI-2027 operating bridge	AI wealth / CAGR	100GW operating bridge	100GW wealth / CAGR
NVDA	R 1,000; m 46%; S 23.5; P/E 22; D 6; C 0	$436.64 / 24.1%	R 2,200; m 45%; S 24; P/E 24; D 8; C 0	$998.00 / 51.5%
AVGO	R 320; m 47%; S 5; P/E 20; D 13; C -2	$612.60 / 17.0%	R 650; m 45%; S 5.2; P/E 22; D 16; C -4	$1,249.50 / 39.0%
VRT	R 36; m 20%; S 0.38; P/E 24; D 2; C 0	$456.74 / 22.0%	R 80; m 21%; S 0.4; P/E 26; D 3; C 0	$1,095.00 / 50.7%
CLS	R 80; m 6.5%; S 0.145; P/E 16; D 0; C 0	$573.79 / 25.6%	R 170; m 6.5%; S 0.16; P/E 17; D 0; C 0	$1,174.06 / 49.3%
NVT	R 12; m 18%; S 0.16; P/E 22; D 4; C 0	$301.00 / 26.3%	R 25; m 18%; S 0.175; P/E 24; D 5; C 0	$622.14 / 50.5%
NRG	EBITDA 7.2; EV 8x; claims 14; S 0.19; D 9; FCF 3.9	$238.47 / 24.1%	EBITDA 10; EV 9x; claims 18; S 0.22; D 10; FCF 5.2	$337.27 / 35.0%
VST	EBITDA 12; EV 9x; claims 20; S 0.3; D 5; FCF 6	$298.33 / 21.3%	EBITDA 18; EV 10x; claims 25; S 0.32; D 6; FCF 8.5	$490.38 / 36.8%
COHR	R 24; m 19%; S 0.23; P/E 20; D 0; C 0	$396.52 / 14.6%	R 50; m 20%; S 0.26; P/E 22; D 0; C 0	$846.15 / 37.7%
TSM	R 390; m 46%; S 5.2; P/E 21; D 20; C 0	$744.50 / 17.0%	R 700; m 45%; S 5.4; P/E 24; D 25; C 0	$1,425.00 / 36.8%
MU	R 360; m 46%; S 1.2; P/E 10; D 8; C 180	$1,568.00 / 20.1%	R 700; m 44%; S 1.3; P/E 9; D 12; C 250	$2,394.31 / 33.1%
ASML	R 105; m 38%; S 0.37; P/E 28; D 40; C 0	$3,059.46 / 18.5%	R 180; m 39%; S 0.39; P/E 30; D 50; C 0	$5,450.00 / 36.3%
SNPS	R 18; m 31%; S 0.2; P/E 30; D 0; C 0	$837.00 / 15.1%	R 28; m 32%; S 0.22; P/E 32; D 0; C 0	$1,303.27 / 28.1%
CEG	R 58; m 14%; S 0.35; P/E 20; D 12; C 0	$476.00 / 14.1%	R 90; m 15%; S 0.38; P/E 23; D 15; C 0	$832.11 / 30.5%
ETN	R 55; m 19%; S 0.38; P/E 24; D 22; C 0	$682.00 / 16.7%	R 90; m 20%; S 0.4; P/E 27; D 26; C 0	$1,241.00 / 34.9%
CRDO	R 8; m 35%; S 0.23; P/E 22; D 0; C 0	$267.83 / 18.4%	R 18; m 36%; S 0.27; P/E 24; D 0; C 0	$576.00 / 42.5%
PWR	R 72; m 8.5%; S 0.15; P/E 25; D 3; C 0	$1,023.00 / 14.1%	R 130; m 8.5%; S 0.17; P/E 27; D 4; C 0	$1,759.00 / 30.1%
KTOS	R 4.6; m 8.5%; S 0.24; P/E 28; D 0; C 0	$45.62 / 4.7%	R 6.5; m 10%; S 0.27; P/E 30; D 0; C 0	$72.22 / 17.0%
MP	R 2.4; m 23%; S 0.215; P/E 24; D 0; C 0	$61.62 / 12.6%	R 4.5; m 25%; S 0.25; P/E 26; D 0; C 0	$117.00 / 31.4%
BNTX	Net cash 10 + pipeline 20; S 0.25	$120.00 / 7.9%	Net cash 10 + pipeline 25; S 0.25	$140.00 / 12.0%
SNDK	R 65; m 40%; S 0.14; P/E 9; D 0; C 120	$1,791.43 / 18.5%	R 130; m 40%; S 0.16; P/E 8; D 0; C 180	$2,780.00 / 31.8%
RELX	R 18; m 27%; S 1.75; P/E 24; D 5; C 0	$71.65 / 15.1%	R 21; m 28%; S 1.78; P/E 23; D 6; C 0	$81.98 / 18.9%

Ticker	AI-2027 operating bridge	AI wealth / CAGR	100GW operating bridge	100GW wealth / CAGR
VRSK	R 4.8; m 34%; S 0.12; P/E 28; D 9; C 0	$389.80 / 17.4%	R 5.5; m 35%; S 0.125; P/E 27; D 10; C 0	$425.80 / 19.9%
MSI	R 21; m 23%; S 0.16; P/E 26; D 24; C 0	$808.88 / 10.8%	R 26; m 24%; S 0.17; P/E 25; D 28; C 0	$945.65 / 15.1%
GEN	R 6.5; m 24%; S 0.61; P/E 14; D 2.5; C 0	$38.30 / 6.2%	R 7.5; m 24%; S 0.64; P/E 13; D 3; C 0	$39.56 / 7.1%
LRCX	R 56; m 35%; S 1.2; P/E 23; D 7; C 0	$382.67 / 14.1%	R 95; m 36%; S 1.25; P/E 25; D 9; C 0	$693.00 / 31.7%

Race, slowdown and ownership: the quoted AI-2027 stock returns require functioning equity ownership and a commercially investable continuation of the capability/compute path. They are not the return in the literal catastrophic race ending. Expropriation, destruction of institutions or loss of human control can make ordinary terminal-P/E analysis irrelevant; a zero-recovery branch is −100%, not a high supplier CAGR. The goals/security forecasts make that distinction more important, not less. AI-2027 race ending, goals forecast, security forecast.

A governed frontier-training slowdown could preserve inference and some existing power revenue while interrupting new chip/tool orders. For an illustrative differential sensitivity, reduce ASML's AI-path earnings 30% and its multiple 20%; reduce NRG's AI EBITDA 15% and EV multiple 10%, leaving claims fixed; reduce VRSK's AI revenue 5%, use a 33% common margin and cut its P/E 10%. These are my stress assumptions, not the authors' probabilities or company guidance. They hurt new capacity suppliers much more than recurring data revenue; nevertheless, Verisk is not protection against institutional collapse. Export controls can additionally impair both foundry utilization and equipment customers; government mobilization can increase volumes while capping profits. The common zero-recovery sensitivity in §6 is deliberately separate from these differentiated commercial effects.

Names with modest returns even on the strong AI path—GEN, BNTX, KTOS and MSI—do not earn capital simply by occupying a socially important category. VRT/NVT/CLS/NVDA have much larger 100GW sensitivities than ordinary expected returns: they are strong physical-buildout bets, not low-risk compounders. A better forecast of AI progress alone does not eliminate that distinction.

3. Their Picks vs Mine: Same Bottleneck and Capital Competition

S01–S04 compare genuinely related economic opportunities. S04 is a contest between my preferred **excluded** equipment challenger and Claude's held equipment name; I did not own LRCX in the original recommendation. S03 covers different interconnect products and distances, not perfectly interchangeable suppliers. S05–S08 are explicitly cross-sleeve capital contests, included to explain the exact allocation changes rather than falsely calling unrelated companies substitutes.

All returns below use the same new underwriting and November 9 prices. “Attractive” means a favorable modeled return with identifiable risk, not a historical valuation percentile that I have not verified. “Fair” means plausible economics without a clear enough advantage to force ownership. “Speculative” denotes especially uncertain financing, clinical/production outcomes or weak owner-cash conversion. These classifications do not imply precise intrinsic values.

Row / bottleneck or explicitly labeled capital contest	My pick / challenger	Their pick / other candidate	Expected CAGR each	AI-2027 CAGR each	100GW CAGR each	Valuation status each	Better stock	Why
S01 Electrical distribution	NVT	ETN	19.4% / 12.6%	26.3% / 16.7%	50.5% / 34.9%	Attractive / Fair	NVT	Lower paid multiple; profitable electrical expansion, not simply a cooling market-share contest.
S02 Existing generation / retail	NRG	CEG	20.1% / 10.4%	24.1% / 14.1%	35.0% / 30.5%	Attractive, levered / Fair	NRG	Returns still win after asset-sale earnings haircut and explicit debt funding; nuclear quality alone is insufficient.
S03 Interconnect	COHR	CRDO	8.8% / 12.2%	14.6% / 18.4%	37.7% / 42.5%	Speculative / Fair, execution-sensitive	CRDO within sleeve; neither held	Credo wins versus Coherent after compensation charges, but loses portfolio capital to MU/VRSK. Different distances/products, not exact substitutes.
S04 Semiconductor equipment preference	LRCX	ASML	8.4% / 12.3%	14.1% / 18.5%	31.7% / 36.3%	Fair-low return / Fair	ASML; neither held	My original preferred excluded WFE challenger loses to lithography after capacity/earnings re-underwriting. Both were not held by me.
S05 Marginal AI-supplier capital, NOT same bottleneck	COHR	MU	8.8% / 12.2%	14.6% / 20.1%	37.7% / 33.1%	Speculative / Fair, cyclical	MU	Higher E and AI returns; less exposure to optical expansion financed ahead of cash receipts.
S06 Marginal profitable-growth capital, NOT same bottleneck	AVGO	VRSK	12.1% / 15.0%	17.0% / 17.4%	39.0% / 19.9%	Fair / Attractive	VRSK	VRSK is my overlooked candidate, not Claude’s pick. Expected advantage exceeds 2pp, and AI return is also slightly higher.
S07 Last capital slot: upstream alternatives	ASML	MU	12.3% / 12.2%	18.5% / 20.1%	36.3% / 33.1%	Fair / Fair, cyclical	MU	Expected returns essentially equal; apply required higher-AI-CAGR tie-break. Neither is a hedge.
S08 Contracted data vs speculative hedge sleeve	VRSK	BNTX	15.0% / 4.3%	17.4% / 7.9%	19.9% / 12.0%	Attractive / Speculative	VRSK	Not technological substitutes: tested as competing capital uses. Return and modeled bust both favor data cash flow.

The electrical comparison is not “Eaton bad, nVent good.” NVT's current guide-based adjusted P/E is about 21×, versus roughly 26× for ETN, using the respective disclosed 2026 outlook midpoints rather than unmatched consensus years. That lower paid valuation and stronger operating growth compensate for some quality and integration risk. For power, CEG's improved guidance deserves a higher estimate, but the stock's higher November 9 price also matters. A superior nuclear fleet need not deliver a superior common-stock return.

The final eight percentage points are genuinely contested. ASML, MU, CRDO, AVGO and TSM cluster near 12% expected CAGR; RELX is around 13%. Within the prompt's approximately 2pp band, MU's 20.1% AI-2027 CAGR beats these alternatives. That is the specified tie-break, not a claim that a few basis points of modeled expected return are knowable. VRSK first earns 15% on a clearer expected-return advantage over AVGO; I do not award it a permanent “data sleeve.”

4. Candidates Only One of Us Found

I distinguish **never considered** from **considered but not held**. Searching the actual deep-dive candidate coverage shows that SNDK is the additional fully underwritten Claude candidate absent from mine. The four fully underwritten Codex names absent from Claude's report are RELX, VRSK, MSI and GEN. BNTX, NVT, NRG, ASML and CRDO are not newly discovered merely because their weights or valuation coverage differ.

Ticker	Discovery	Price + date	Expected CAGR	AI-2027 CAGR	100GW CAGR	Verdict
SNDK	Claude only	$886.51; 2026-11-09	9.5%	18.5%	31.8%	Reject: MU wins on expected and AI return. Useful omission to correct, not a new holding.
RELX	Codex only	$40.02; 2026-11-09	13.3%	15.1%	18.9%	Reject at margin: VRSK wins E and AI; MU wins AI tie-break versus RELX.
VRSK	Codex only	$200.60; 2026-11-09	15.0%	17.4%	19.9%	Add 15%; replace AVGO, via H20/S06. No diversification entitlement.
MSI	Codex only	$527.97; 2026-11-09	7.6%	10.8%	15.1%	Reject: modest E and AI return; no demonstrated disaster payout.
GEN	Codex only	$29.80; 2026-11-09	5.3%	6.2%	7.1%	Reject: low paid multiple does not overcome low common-earnings growth.

SNDK is a useful correction to my research universe, but it does not win capital. Its NAND-heavy economics are not identical to qualified HBM, and terminal normalization still matters. All SanDisk prices, EPS, shares and wealth values here are on the **pre-8-for-1-split** basis appropriate to November 9; the record date is not the split-adjusted trading date. Dividing price/EPS/wealth by eight and multiplying shares by eight must leave every return unchanged. F19 records the timing and cash conversion evidence.

RELX, VRSK, MSI and GEN retain their original Codex operating and valuation assumptions; only prices change. I did not inflate Verisk's model to justify the new weight. Its ranking rises because competing forecasts fall. Verisk's data and workflow economics merit capital only while they support the modeled revenue, margin and cash conversion. Neither ownership of information nor a public-safety/privacy label guarantees an enduring bottleneck. Confidence is moderate for VRSK/RELX cash-flow durability and lower for forecasting how quickly AI changes their distribution and pricing.

5. Facts That Change a Pick

The table separates verified company disclosures, source-scenario assumptions and my valuation judgments. Dates are publication dates unless explicitly labeled. Older inputs are marked STALE (>30 days) where used as current operating anchors. I do not treat source authority as verification of a future outcome. Repeated references to F-numbers above point here; the numerical ledgers are my models, not quoted forecasts.

#	Claim	Mine	Theirs	Primary-source check: source + date	Resolved	Pick affected
F01	NRG's new outlook and asset sale can fund both debt reduction and growth without losing earnings.	Preferred NRG, but my original terminal bridge needs updating for the newly announced sale.	Dismissed NRG's mix without a complete terminal model.	NRG Q3 release, Nov 9; asset-sale announcement, Nov 9. 2027 EBITDA guide $5.5–6.0bn and FCF-before-growth $2.9–3.4bn exclude the planned 4.9GW sale; proceeds $4bn, closing targeted Q2 2027.	Remove sold earnings and allocate proceeds once. My $0.5bn earnings subtraction is an assumption, not disclosed asset EBITDA. Terminal $14bn claims require substantial repayment from current roughly $22.8bn net financial claims.	NRG retained over CEG, with a stricter bridge.
F02	CEG's 2026 earnings anchor is unchanged.	Older midpoint about $12.00.	Older guide also underlies the comparison.	Constellation Q3 release, Nov 9: adjusted EPS guidance rises to $12.50–12.80.	Update the operating anchor, but also reprice to $276.09. An earnings upgrade does not override the return comparison.	CEG still zero; its new expected return is 10.4%.
F03	VST's 2027 guide excludes Cogentrix, allowing an additional acquisition uplift.	Used consolidated enterprise economics.	Described the guide as excluding Cogentrix/Meta, then added an increment.	Vistra earnings release, Nov 6: consolidated 2027 EBITDA $7.7–8.5bn and FCF-before-growth $4.3–5.1bn, following the October Cogentrix close.	The cited release provides no basis for adding another $0.7bn acquisition uplift on top. Meta's exact incremental contribution is not separately established here. FCF before growth is not entirely distributable.	Retain VST, lower my expected return and reject unsupported double counting.
F04	NVT's smaller cooling share makes ETN the better investment.	Own NVT for electrical/thermal economics and paid valuation.	Preferred ETN and treated cooling-share limits as disqualifying.	nVent SEC release, Oct 30: Q3 sales +44%, also +44% organic; FY adjusted EPS $5.36–5.39. Maverick closed October 1, after that quarter. Eaton release, Nov 3.	Acquisition growth does not explain away NVT's reported organic growth. Compare entire electrical businesses at current prices, while retaining integration and execution discounts.	NVT 12%; ETN 0%.
F05	CLS has insufficient shareholder economics to justify a position.	Held 12%, but with too little dilution allowance.	Rejected at a $378 base terminal wealth.	Celestica SEC release, Oct 26: 2027 revenue outlook $40bn, adjusted EPS $21.50, adjusted operating margin 9%, capex $1.5–2bn; FY26 FCF guide $460m.	Use the issuer outlook as a growth anchor, not as GAAP owner EPS. Base 2030 revenue $65bn, normalized common margin 6.5%, 145m shares and 16× allow profit growth while charging expansion.	CLS 12% remains superior to exclusion.
F06	Credo's adjusted EPS can be capitalized without a substantial compensation/dilution charge.	My growth case was too low.	Higher EPS case, including a cited CY28 consensus figure not independently verified here.	Credo Q1 FY27 release, Sep 1 — STALE: $479m revenue; GAAP/non-GAAP EPS $0.67/$1.20; stock compensation $87.979m, about 18.4% of revenue.	Raise revenue, but use common margin after economic compensation and 230m base shares. Do not label the unverified consensus number a confirmed source fact.	CRDO improves to 12.2% expected, still loses MU's AI tie-break.
F07	Broadcom's financing arrangements equal an immediately payable $60bn corporate debt.	Original valuation did not separately reserve this risk.	Correctly emphasized financing exposure, but the headline face amount is not sufficient valuation accounting.	Broadcom 10-Q, Sep 10: initial maximum backstop roughly $29bn. Oct 27 8-K: newer maximum undiscounted exposure about $43bn, tied to 72.5% of remaining lease payments less realized rack value. Q3 results, Sep 2 — STALE: strong revenue and cash conversion remain real.	Credit support is contingent and recovery-sensitive. Do not blindly add overlapping transaction headlines or subtract the entire financing face amount from equity. Use explicit reserves and correlated downside; lower the exit premium separately.	AVGO loses to VRSK, not because a $60bn immediate loss is assumed.

#	Claim	Mine	Theirs	Primary-source check: source + date	Resolved	Pick affected
F08	Micron's memory earnings must rapidly revert to the old cycle despite contracts.	Original base $80 EPS, explicit cash; normalized too harshly relative to contract evidence.	Higher $130 EPS plus an implicit share-reduction benefit.	Micron Q4 prepared remarks, Sep 30 — STALE: 26 strategic agreements representing an estimated >35% of revenue through 2030; most have defined pricing frameworks, often floors/ceilings. Customer deposits are financing, not operating cash.	Raise durable earnings but do not assume locked peak margins. Use $110 EPS, 10×, fixed shares and explicit net cash after deposit claims. Company-wide quarterly revenue is not an HBM-only TAM observation.	MU replaces COHR.
F09	ASML's nearer-term supply plan supports only my old low terminal EPS.	Underestimated the revenue/margin opportunity.	Closer on EPS and chose an 8% allocation.	ASML Q3 results, Oct 14: FY26 sales €44–45bn; plans to raise 2027 low-NA EUV/DUV capacity about 30%, with further expansion considered for 2028.	Rebuild units, mix, service, FX and shares; do not equate capacity with sales. The resulting base EPS is $85.65, close to Claude's direction for independent reasons.	ASML beats LRCX but loses the final slot to MU.
F10	AI-2040's roughly $1.2m “per wafer” is a foundry wafer ASP that supports TSM rents.	Earlier audit distinguished full-system cost from foundry revenue.	Used the headline in a way that overstates the direct pricing implication for TSM.	AI-2040 compute supplement, accessed Nov 9: its all-in wafer-equivalent cost allocates much more than foundry processing; the stated foundry component is roughly $25–30k.	Supplier markups, HBM/packaging and deployment infrastructure are not all TSM revenue. My new base revenue rises, but I do not give TSM a foundry-price uplift or favorable probabilities from this unit mistake.	TSM stays zero; its earnings opportunity is real but the argument for extra rents is weaker.
F11	EUV/HBM scarcity and a financing model prove memory cannot glut or investment cannot pause.	Treated them as conditional mechanisms.	Stronger confidence in scarcity and financing conclusions.	Same AI-2040 supplement: HBM uses a large modeled share of EUV passes; the 2028 financing category combines debt and equity. Its scarcity surcharge and later spending pause are assumptions.	Neither an assumed cost surcharge nor financing mix is a supplier profit guarantee. HBM resource intensity can slow expansion, not abolish substitution or cyclicality.	MU gets a cyclical multiple; ASML gets capacity upside, not automatic surcharge pass-through.
F12	The older spreadsheet's dollar outputs are a current, directly backtested HBM forecast.	Prior local audit recomputed formulas and bottleneck minima.	Compared some memory/compute headlines without fully reconciling categories.	Shared chip-compute workbook, cached values/formulas audited Nov 9: high-demand conditional model; HBM is the minimum through 2030. The old $6.69bn anchor and cumulatively increasing price exponents materially affect its $306.4bn 2030 TAM.	Constant 3% annual price growth instead gives about $164.7bn with the same old quantity/anchor assumptions. Neither is a contemporary calibrated TAM. Do not compare total Micron sales with HBM-only dollars.	Raises attention to MU's supply constraint without importing false precision into valuation.
F13	AI-2027 has been “70–90% accurate,” so the exact 2027 endpoint deserves that probability.	Audit separated pace, accuracy and updated timelines.	More favorable narrative interpretation of forecast validation.	AI Futures timeline update, Aug 16 — STALE as a current calibration; published compute forecast; current model framework. The update's pace comparison is not a calibrated hit rate.	Give rapid progress substantial weight, but distinguish updated coder/research milestones from exact-year transformative AI. Leading-lab recognized revenue, global ARR, annual capex and the value of installed compute are different variables.	AI-like commercial expansion remains most likely; no arbitrary 70–90% company bull probability.

#	Claim	Mine	Theirs	Primary-source check: source + date	Resolved	Pick affected
F14	A 100GW national buildout across many campuses equals the required single training cluster.	Original discussion included the single-cluster challenge but also used broader buildout language.	Described a multi-campus realization that is easier than the new named definition.	Situational Awareness chapter, June 2024 PDF, Tables 4–5: largest-cluster power and worldwide annual investment are distinct.	Model one 100GW training cluster plus the associated much larger global industry. Keep the path low-probability and identify power, network, permitting, chip/memory and funding requirements.	Raises conditional physical-supplier upside, but does not make the extreme path the expected-value case.
F15	Kratos procurement visibility supports a premium on adjusted earnings despite investment.	Excluded after owner-earnings/dilution charge.	Held 2%, treating it partly as strategic resilience.	Kratos SEC results, Nov 5: 2027 revenue $2.1–2.2bn and EBITDA $230–240m; 2026 FCF usage $110–120m; Q3 GAAP EPS zero versus adjusted $0.24.	Funded demand is valuable, but cash-consuming production and dilution stand between revenue and common returns. A crisis does not automatically make the equity a hedge.	KTOS remains zero; weakest Claude holding.
F16	BioNTech's current cash guarantees a high 2030 downside floor.	Considered but not fully valued in the original; no holding.	$65 bear floor and 1% hedge-like allocation.	BioNTech Q3 update, Nov 3, not Nov 9: cash/securities €16.164bn, about 247.9m shares, continuing losses.	Cash today is not cash after four years of R&D and operating burn. Fixed $1.10/€ is a valuation assumption, not claimed spot FX. Use post-burn cash and risk-adjusted pipeline, with no duplicated partnership proceeds.	BNTX remains zero at 4.3% expected CAGR.
F17	Coherent's adjusted EPS improvement adequately funds the optical expansion.	Held 8%, too favorable a margin/share/multiple combination.	Held 5%, more cautious valuation but still allocated.	Coherent SEC release, Nov 4: revenue $2.37bn, adjusted EPS $2.09; operating cash flow −$92.7m and capex $775.5m.	Cash timing and expansion costs matter. The new model still allows large growth; it lowers the share of that growth captured by each current share.	Both old weights are too high relative to MU; mine goes to zero.
F18	Verisk was too low-return to compete for concentrated capital.	Fully modeled but excluded at the original opportunity cost.	Not considered.	Verisk SEC release, Nov 5: FY revenue guide $3.190–3.240bn, adjusted EPS $7.45–7.75; Q3 organic constant-currency growth 7.4%. Nine-month FCF was about $824m while buybacks were much larger; debt and timing effects matter.	The original long-run model remains plausible, but recent buybacks cannot be perpetuated without funding. Its relative rank improves after other models are corrected. At $200.60, the entry is about 26.4× guide-midpoint adjusted EPS, not a demonstrated historical bargain.	VRSK replaces AVGO, conditional on revenue durability, margins and cash.
F19	SanDisk's large EPS and upcoming split imply a valuation mismatch.	Did not underwrite it.	Underwrote but excluded.	SanDisk FY27 Q1 release, Oct 29: revenue $11.031bn, adjusted EPS $49.51, adjusted FCF $5.696bn; 8-for-1 split-adjusted trading begins Nov 20 after a Nov 9 record date.	November 9 remains pre-split. Normalize NAND earnings and cash, including JV/investment obligations; no eightfold price/EPS mismatch.	Add to researched universe, not allocation.
F20	Lam's equipment growth deserves my original valuation despite cyclicality and China exposure.	Preferred excluded WFE challenger.	More cautious, preferred ASML.	Lam September-quarter results, Oct 21: revenue $8.22bn, GAAP/adjusted EPS $2.19/$2.20, China 35% of revenue.	Geographic concentration is verified; a new hypothetical ban is not. Lower revenue/margin/multiple assumptions for risk without fabricating an enacted restriction.	ASML wins the WFE comparison.

One additional TSM operating anchor is important: TSMC's October 15 quarterly release reports a 51.1% net margin and the next-quarter revenue outlook. That supports taking the earnings opportunity seriously; it does not establish a permanent 51% margin. My 46% base common margin, 21× terminal multiple and symmetric 25/50/25 financial-state probabilities deliberately separate stronger operating evidence from the incorrect all-in-wafer pricing argument.

Market-price provenance: research/codex_cross_critique/quotes.json and the corresponding raw Yahoo chart responses preserve the November 9 close used for every row. These are public vendor prices, not exchange-certified executions. Spot checks included NVDA historical prices and NRG historical prices. An after-hours quote or a later split-adjusted series must not silently replace one side of a comparison.

6. Revised Allocation: Codex After Critique

This is my revised Step 2 recommendation within the contested and one-model-only candidate set. It is neither an average of the two portfolios nor the final all-universe Step 3 ranking. The recommendation keeps the original 15% default position ceiling, lower 12% execution/leverage limits for CLS/NVT/NRG, and 23% combined NRG/VST exposure. Those are judgmental risk constraints, not an optimizer's discovery or a claim that downside is small. The best-ranked positions fill their limits; VRSK earns 15%, and MU wins the remaining 8% through the required AI-return tie-break.

Ticker	Original %	Revised %	Delta pp	Head-to-head justification
NVDA	15.0	15.0	+0.0	H01: return advantage survives lower multiple; keep cap.
AVGO	15.0	0.0	-15.0	H02 + H20 + S06: VRSK wins the full 15-point capital contest.
VRT	15.0	15.0	+0.0	H03: lower forecast, still wins its funded slot; keep cap.
CLS	12.0	12.0	+0.0	H04: current guidance supports higher value than Claude's terminal hurdle.
NVT	12.0	12.0	+0.0	H05 + S01: beats ETN after integration/dilution allowance.
NRG	12.0	12.0	+0.0	H06 + S02: beats CEG after full asset-sale/debt bridge.
VST	11.0	11.0	+0.0	H07: power return wins, combined power cap remains 23%.
COHR	8.0	0.0	-8.0	H08 + H10 + S05: all 8 points to MU.
VRSK	0.0	15.0	+15.0	H20 + S06: all 15 points from AVGO.
MU	0.0	8.0	+8.0	H10 + S05/S07: wins COHR replacement and AI tie-break.
TOTAL	100.0	100.0	0.0	100% common stocks; maximum 15%; no external funding.

Checksum: 15 + 15 + 15 + 12 + 12 + 12 + 11 + 8 = **100.0%**. Funding checksum: AVGO −15 and COHR −8 fund VRSK +15 and MU +8; net change **0.0 percentage points**. All other Claude-only holdings remain zero. All positions are listed common shares/ADRs; no options, leverage, shorts, private exposure or new hedge sleeve. Event Horizon remains approximately 50% of household net investments; Foundation and Continuity are unchanged and Moonshot remains 0%. No trades were executed.

Apples-to-apples portfolio results

Every row below uses the **same November 9 prices and same new stock models**. The Claude-weight diagnostic asks what its allocation produces under my adopted inputs; it is not presented as Claude's own forecast and is not a blended recommendation.

Portfolio, SAME new underwriting	Expected CAGR	AI-2027 CAGR	100GW CAGR	2030 bust total return	2030 delay total return	HHI
Codex original, new underwriting	17.3%	22.3%	44.9%	-65.0%	-28.9%	0.1292
Codex revised	17.9%	22.7%	42.5%	-57.9%	-21.6%	0.1292
Claude weights, new underwriting	13.7%	18.5%	38.8%	-64.9%	-33.2%	0.0852

The expected terminal total return rises from 93.7% to 97.9%. AI-2027 CAGR improves approximately 0.4pp; 100GW CAGR falls approximately 2.4pp. The extreme-path sacrifice is deliberate: Verisk has less physical-capex torque than Broadcom, while MU beats Coherent on the primary comparison. Expected terminal wealth, not maximum scenario upside, governs the decision.

Price, underwriting and allocation attribution: my original published expected CAGR was about 22.8%. Holding its terminal assumptions fixed but using November 9 prices and the shorter horizon gives 20.9%. Adopting the new operating/valuation assumptions with the original weights reduces it to 17.3%. Changing only the weights then raises it to 17.9%. Claude's frozen original models at the same new prices imply about 13.2%; that is different from the 13.7% diagnostic under my new models. None of these changes is an observed investment return.

Comparable busts, not incomparable labels: Claude's published approximately −26.6% stress combines bear values for most names with base values for KTOS/MP/BNTX. Putting every name at its own original bear value produces approximately −31.8% on the old prices; at November 9 prices those two figures become approximately −30.5% and −35.3%. That roughly five-point difference comes from scenario assignment, not demonstrated hedge performance. Under the new common economic stress, Claude's weights lose about 64.9%, my original weights 65.0%, and the revision 57.9%. These are model endpoints, not empirical maximum drawdowns or loss quantiles. They cannot establish which portfolio is actually safer in every crisis.

The delay stress interpolates 40% of operating progress from bear toward base, then reduces the interpolated valuation multiple by 10%, while preserving the corresponding shares and claims. For BNTX it interpolates SOTP wealth. It is a consistent sensitivity, not a fifth state secretly added to expected value. A severe interim drawdown, liquidity event or common zero-recovery regime can be worse than either terminal stress.

Fragility and ownership-risk sensitivity

Assumption change	Original weights CAGR	Revised weights CAGR
Central comparison	17.3%	17.9%
VRSK: every terminal P/E 20% lower	17.3%	17.2%
AVGO: every terminal P/E 20% higher, other inputs unchanged	18.0%	17.9%
Extra common 10% zero-recovery regime	14.4%	15.0%
Extra common 20% zero-recovery regime	11.1%	11.7%

The multiple tests change only the selected valuation component; dividends, cash, other stocks and scenario probabilities stay fixed. The zero-recovery tests place an **extra illustrative** common 10% or 20% state outside the commercial model, with all modeled terminal wealth in that state set to zero. They are not asserted catastrophe probabilities and are not folded into the reported primary expected return. If some distributions survive, recovery would be higher; if institutions cease functioning, ordinary financial utility may cease to be the relevant objective.

The practical conclusion is narrower than “VRSK is certainly superior.” It is that the evidence supports replacing a more aggressively valued, financing-exposed supplier with a previously researched cash-flow business **under these explicit assumptions**. A relatively small change in the terminal multiple can reverse that judgment. The revised portfolio still has 62% in semiconductor/thermal/electrical integration exposures, 23% in merchant/retail generation and 15% in data/workflow. These categories share customers, financing conditions and economic activity; they are not independent risk factors. Top-three weight remains 45%, and HHI remains 0.1292 despite the improved modeled bust.

At roughly half of household net investments, a 57.9% Event Horizon terminal loss alone would subtract about 29% of household investment value before any change in the other portfolios. Continuity can fall at the same time. This is a substantial accepted risk, not a newly established household loss tolerance. The mandate still favors expected return; it does not supply permission to describe this downside as safe.

Capital opportunity costs and reversal gates

Retained/added holding	Why it wins capital now	If removed, the next comparison	Evidence that would reverse the decision
NVDA 15%	Highest expected return among the funded candidates even after a lower exit premium.	MU/ASML/CRDO after respecting existing risk limits; then the broader Step 3 universe.	Customer economics or alternative accelerators lower durable common margins/market share enough to erase the return lead.
NRG 12%	Beats CEG after sale earnings are removed and debt funding is explicit.	VST only within the power cap; otherwise MU/ASML.	Failed sale, weak post-sale cash, higher claims, or new-build obligations that break the funding bridge.
NVT 12%	Stronger expected return than ETN at a lower guide-based paid multiple.	ETN for electrical exposure, but MU/ASML can win cross-sleeve capital.	Organic growth and cash conversion decelerate while acquisition/dilution costs rise.
CLS 12%	Current issuer outlook supports the earnings bridge; 16× does not require a premium software multiple.	MU/ASML/CRDO, not an automatic replacement integrator.	Customer concentration, lower common margins, capex overruns or share issuance overwhelm sales growth.
VST 11%	Existing-generation economics beat CEG; no acquisition double count required.	NRG only if its separate cap is justified upward; otherwise upstream suppliers.	FCF-before-growth fails to become distributable cash after mandatory investment and financing.
VRT 15%	Revised 16.0% expected CAGR remains above the close-return challenger group.	VRSK only if its cap changes; otherwise MU/ASML/CRDO.	Lower backlog conversion, margin normalization or required equity funding undermines the $15 base EPS.
VRSK 15%	Beats AVGO by about 2.9pp expected CAGR, with a slightly better AI-path return.	AVGO and RELX; broader platform challengers must be checked in Step 3.	License growth, common margin or owner-cash conversion misses; debt-funded repurchases persist; agents substitute away from paid data/workflows.
MU 8%	Beats COHR outright and wins the AI tie-break versus the near-12% challenger group.	ASML first on current expected return; CRDO/AVGO/TSM remain close.	Contract pricing fails to cushion normalization, deposits are miscounted as cash, or capacity growth overwhelms qualified demand.

These are review conditions, not automatic orders. Account-specific capital-gains taxes, spreads and household overlap can alter implementation; they were not provided for this head-to-head and are not invented. The existing deployment plan should not be treated as silently updated by this critique.

7. Picks Still Contested

Ranked by the approximate amount of capital and valuation uncertainty affected, not by how emphatically either report defended the thesis. “Weight at stake” is exposure under debate, not a pre-authorized trade size.

Rank	Ticker or bottleneck	My position after critique	Claude deep-dive position	Weight at stake	Evidence that would settle it
1	VRSK versus AVGO; remaining platform opportunity cost	VRSK 15%, AVGO 0%; old Verisk model survives, Broadcom model reduced.	AVGO 6%; VRSK absent.	Entire 15% replacement, plus broader Step 3 alternatives.	Verisk renewal/organic growth and cash after compensation, dividends and debt; Broadcom counterparty/recovery disclosures and sustained common cash conversion; uniform repricing of previously underwritten non-contested platforms. A 20% multiple test already shows this is not settled.
2	Merchant/retail power versus nuclear premium	NRG 12%, VST 11%, CEG 0%.	VST 9%, CEG 6%, NRG 0%.	23% own power exposure; composition materially different.	NRG sale close, disclosed asset earnings and actual post-sale cash/debt; consolidated VST guide scope and required growth spending; realized contract returns at CEG. Financing and affordability constraints can dominate the headline GW demand.
3	NVDA terminal economics and common demand factor	15%, with lower premium than my original.	13%, lower expected return.	15% direct, with effects across suppliers.	Multi-year paid utilization, customer cash economics, retained share versus custom accelerators, export restrictions and common margins after new product transitions.
4	CLS scale-to-cash conversion	12%.	0%.	12%.	Conversion of the 2027 outlook into collected cash after working capital and capex; share count and common margin rather than adjusted EPS alone.
5	NVT versus ETN	NVT 12%, ETN 0%.	ETN 5%, NVT 0%.	12% electrical allocation.	Comparable organic growth, acquired-business returns, margins and free cash after integration; customer evidence that delivery scarcity persists.
6	VRT exit premium and cash-funded capacity	15%, lower valuation.	9%.	6pp weight difference; 15% direct exposure.	Sustained profit-to-cash conversion and service/installed-base economics sufficient to support the modeled multiple after order growth normalizes.
7	MU versus ASML/CRDO/TSM	MU 8%; others zero despite improved ASML/CRDO estimates.	MU 9%, ASML 8%, CRDO 3%, TSM 14%.	Last 8% own slot; much larger Claude upstream basket.	Memory contract realization and deposit-adjusted cash; EUV accepted deliveries/service mix; Credo compensation-adjusted earnings; foundry margins and Taiwan exposure. Small expected-return gaps make the AI tie-break decisive today.
8	COHR optical expansion	0%, down from 8%.	5%.	Former 8% slot.	Sustained positive operating cash and fully financed capacity, with common margin/share count sufficient to beat MU rather than merely validate optical demand.
9	Small defense/materials/biotech sleeve	KTOS/MP/BNTX all zero.	Combined 5%.	5% of Claude's capital.	Funded production converted to positive owner cash; probability-adjusted clinical value; evidence of actual crisis payoffs rather than assigning base values in a stress table.

Verification and reproducibility

The supporting folder is research/codex_cross_critique/. fetch_quotes.py and raw responses preserve price provenance; critique_model.py contains frozen original endpoints, all new operating inputs, named paths, reconstruction rules and portfolio math; critique_results.json preserves full precision. sync_tables.py inserts generated numerical tables so the report does not depend on manually copied returns. The accompanying verification script checks price dates, coverage, probabilities, wealth calculations, allocation and funding checksums, scenario completeness and table synchronization.

Scope check: 19 required portfolio disagreements, one additional capital-winning head-to-head, 25 fully modeled names, five one-model-only discoveries, and three return metrics for every candidate in §§2–4. Both original deep dives remain unchanged; Claude's separate Step 2 critique was not consulted. No commit, push, deployment edit or trade is part of this run.